Math 464 Capstone Project

The capstone project for Math 464 will consist of three parts. This project will be due on **Sunday, May 3rd at Midnight**. I cannot accept late work on this project, since there are a large number of students in the course and I have tight deadlines to finish grading before final grades are due. **Be sure to start early and save often**.

Part 1 (Implementation)

Implement the revised simplex method and the tableau simplex method. Your implementation should allow you to use Bland's Rule or the Lexicographic pivoting rules. You implementation should take in an LP in general form (with some collection of inequality and equality constraints) and should convert it to standard form before running the appropriate version of the simplex method. Your implementation should also find an initial BFS using Big-M or the Two-Phase Simplex method. **You are allowed to use standard Python libraries (like Numpy), but you are not allowed to use a solver in this part.**

Part 2 (Application)

Model a real world problem as an LP. State clearly what the problem is and detail your formulation. Find (or create) data for this real world problem. Create at least 3 data sets of various sizes.

Part 3 (Report)

Take both of your implementations of the simplex method (from part 1) and use them to solve the real world problem (from part 2). Also use a standard solver (such as PULP) to solve the real world problem. Compare the runtimes of both your implementation and the solver. You should include plots comparing the runtimes to the size of the datasets. You should track the number of decision variables as well as the number of constraints when comparing the dataset sizes. Finally, the report should give a brief summary of your findings.

Rubric

Part 1 will be worth 80 points, part 2 will be worth 60 points, and part 3 will be worth 40 points for a total of 180 points. Here are the rubrics I will be using for grading each part:

Part 1				
Score	Accuracy	Comments	Conversion/Initialization	Implementation
20	The code works as expected.	You code is well commented and easy to follow.	Your code correctly converts the LP to standard form. It also initializes the simplex method with a BFS appropriately.	The simplex method is implemented in the variations described.
10	The code mostly works on some cases, but some bugs are present.	A few comments are present, but the code isn't well commented or easy to follow.	Your code has difficulty with initialization or conversion. It will work in some cases, but not all.	The simplex method is implemented for some but not all variations.
0	The code doesn't work at any level, or the simplex method wasn't implemented.	No comments present or comments are irrelevant to code.	Your code doesn't initialize or convert the LP to standard form.	The simplex method is not implemented using standard libraries (i.e. a solver is used or the simplex method is not implemented).

Part 2			
Score	Problem Statement	Problem Modeling	Data
20	The problem is clearly and unambiguously stated. Any simplifying assumptions are clearly stated.	The LP accurately models the problem in the problem statement.	The data was created or collected correctly. The method of creating data is clearly stated and any data collection/cleaning methods are clearly stated. Any collected data is also cited correctly.
10	There are some ambiguities in the problem statement.	The LP model is similar to the problem statement, but modifications are made to the problem statement.	The created/collected data isn't a full 3 datasets. The collection methods are not clear and/or the data creation has some underlying flaw.
0	The problem isn't stated. (Even if an LP is modeled).	The model is not given or is completely different from the problem statement.	The data collection/creation method isn't detailed. Citations are missing and/or data is included with no mention of where the data came from. Data may also be missing.

Part 3		
Score	Solver Implementation	Analysis
20	The solver correctly solves the given LP. The solver is used correctly.	The plots are clearly labeled and contain the relevant details. The report clearly summarizes your findings.
10	The solver implementation contains some flaws.	The report is missing some details or the plots are unclear.
0	The solver isn't used to solve the LP.	No plots or summary is given in the report.